A Molecular Communication System in Blood Vessels for Tumor Detection

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ABSTRACT
This paper shows a proposal of a biological nano-communication system established in a blood vessel, aiming to support the detection and treatment of tumors. This system could either be used for diagnostic purposes in the early stage of a disease or to check any relapse of a previous disease already treated.

In our proposal, the tumor detection happens through revealing tumor biomarkers on the cell surface, such as the CD47 protein. This detection takes advantage of some recent proposal if implementing nanorobot transport systems through modified flagellated bacteria. When a biomarker is detected, a molecular communication system is used for distributing the information over a number of nano-machines. These machines have a size similar to the white blood cells, so that they can flow through the vessel at the speed of the largest particles. The transported information is detected extra-body, through the use of smart probes, which triggers a decision tree in order to estimate the nature of the tumor and its most likely location.

Categories and Subject Descriptors
I.6.3 [**Simulation and Modeling**]: Applications.

General Terms
Design, Performance, Theory.

Keywords
Biological Nano-Communications, Tumor detection, Simulation, Biomarkers.

1. INTRODUCTION
Biological nanoscale communications is a promising research area that can offer significant benefits to a wide range of applications [1][2], including medicine [3]. Research at the nanoscales has been active for decades, producing significant results, among others, in the design of biocompatible nanomachines and nanorobots [4][5]. Nevertheless, only recently the networked coordination of nanomachines has been objects of intense research. This is clearly a still open and intense research area, aiming at establishing effective and reliable communications between entities at the nanoscales. Besides the clear need of revising all established communication models, additional challenges emerge when such communication systems must be bio-compatible and established within human bodies.

In this paper we propose a solution for establishing *digital* transmission, used for exchanging multiple *messages*, between entities in *blood vessels*.

The need of considering this particular communication environment emerges by the current research in the medical fields of cardiocirculatory system and tumor detection and treatment. For example, monitoring the concentrations of biomarkers in the cardiocirculatory system can allow detecting critical diseases in their early stage and release specific medicines by nanorobots released in vessels. A pioneering implementation of nanorobots designed for releasing medicine in tumor sites is illustrated in [4]. Through the use of flagellated bacteria, the authors showed a mechanism for transporting nanorobots in the desired location for releasing a suitable amount of medicine in the appropriate location of tumors. Besides the specific application indicated in [4], which is indeed extremely important, this mechanism could also be exploited even for implementing other actions related to the treatment of tumors, since their detection, which is the object of this paper. In particular, we will show the establishment of a digital communication system, making use of nanorobots transported by flagellated bacteria, referred to as bacteriobots in [4], allowing the implementation of an early tumor detection system.

Metastasis is the spread of a cancer cells to other parts of the body; about 90 percent of cancer deaths are related to metastases. Surgery and radiation are effective at treating primary tumors, but difficulty in detecting metastatic cancer cells has made treatment of the spreading cancers problematic. The blood stream will replace the skin as the interface for sensing and calibrating therapies.

A critical aspect of such a system is the information delivery to the external. In particular, the blood requires about one minute to complete the large and small circulation. Hence, this circulation provides a means to rapidly deliver the information to the desired probe location, and from the probe to the external.

Unfortunately, nanorobots, remaining close to the tumor site, cannot deliver the information that they can detect. For this reason, we resort to a communication system that can transfer this information to other devices, that can be effectively transported

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through the cardiocirculatory system. The analysis of this communication system will demonstrate the effectiveness of the system. To this aim, we will make use of BiNS2, a simulation package for biological nano-networks [6][7] which has been tuned by using experimental results of biological experiments [8]. It allows understanding the communications potentials in blood vessels, along with the effects of biological phenomena of great interest, such as aging. Through the use of parallel simulation techniques, BiNS2 allows simulating multiscale bounded spaces.

The proposed communication system includes mobile transmitters and receivers.

The cardiocirculatory system is a multiscale system, due to the different size of the particles and vessels. Our analysis approach is coherently multiscale, involving tissues, cells, and molecules within vessels having a length up to 3 mm. In particular, our analysis will consider microcirculation, which is the blood delivery to the smallest blood vessels. In fact, this is the type of circulation feeding tumors, and necessarily including the bacteriobots detecting them.

The analysis illustrated shows the capabilities of the nano-communication system to deliver messages encoded by a vector of binary symbols. Each bit is physically transmitted by a burst of particles released by a transmitter and propagating through the vessel. Receivers measure the signal at different locations, both along the vessel wall and in other mobile receivers.

On the basis of the achieved results, we show a simple but effective scheme for implementing a biocompatible digital receiver, able to auto-synchronize and decode the information received by means of repeated pulses of information molecules.

The paper is organized as follows. Section 2 illustrates the medical objective that has stimulated our research. Section 3 illustrates the biological environment where communications are established. The communication models are shown in Section 4. Section 5 illustrates the numerical analysis. Section 6 reports some final comments and research directions.

2. MEDICAL SCENARIO

The medical scenario consists of detecting the presence and progression of a tumor through either the robot convergence towards the tumor site or the detection of tumor cells in the blood stream.

In healthy conditions the immune system can eliminate sporadic tumor cells circulating in blood vessels (circulating tumor cells - CTCs) [24]. Differently, in the presence of a tumor disease, the early detection of tumor cells is essential, either circulating or in the tumor site. Hence, monitoring CTC concentration within vessels is a method for detecting any relapsing of a disease after a treatment considered successful.

CTCs are originated from the primary tumor site. Though the blood stream they can reach and metastasise any part of the body, thus spreading the disease. Thus, the value of CTC concentration in the bloodstream provides an effective diagnostic and prognostic information about location and progression of a tumor. For example, a micro-fluidic chip capable of capturing CTCs with 90% success rate is illustrated in [19]. This result, and similar ones [20][21], have greatly stimulated research activities.

Detection of CTCs happens through revealing the presence of specific biomarkers located on tumor cell surface. In order to gain a deep knowledge about the nature and the location of a tumor, this detection should consider different markers, and use them to drive the estimation process by a computer-based decision tree. Some important biomarkers are illustrated in what follows.

A very important biomarker, which has promoted a significant research effort, is generated through the mechanisms by which tumor cells can avoid macrophages, which are cells having the task of removing dying or dead cells and cellular debris. Tumors try to evade the immune system by exploiting the regulatory mechanisms that protect healthy cells from immune-mediated attacks [22]. In particular, the CD47 protein, an immunoglobulin (Ig)-like receptor, is exposed by many cell types in order to indicate to macrophages that they should not be eliminated. Most of tumors over-expresses CD47. The binding of CD47 expressed on the cell surface with the signal regulatory protein-α (SIRP-α) protein carried by macrophages allows tumor cells to evade the macrophage's mediated immune response.

In simple words, expression of the CD47 by a cell has the effect of communicating to macrophages a "don't eat me" signal [22][25]. In fact, macrophages can recognize CD47 molecules by a signal regulatory protein-α (SIRPα), also known as SHPS1. It acts as a receptor for CD47, thus blocking the phagocytosis of macrophages, as schematically shown in Figure 1.

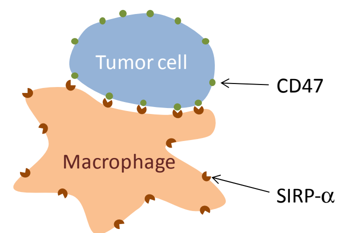

Figure 1. Tumor cell blocking phagocytosis though CD47 exposure.

For this reason blocking this mechanism through modifying the relationship between CD47 and SIRP-α is an active research area [23]. Current nanomedicine approaches consider RNA interference (RNAi) technology by means of liposomes made with protamine-hyaluronic acid and loaded with anti-CD47 siRNA. This has resulted in an efficient silencing of CD47 and tumor regression [13].

It is worth to mention other important biomarkers, such as the circulating microRNA-101 for the hepatocellular carcinoma [32], the plectin, for the pancreatic cancer [33], the apolipoprotein C-II for the cervical cancer [34], the CD164 protein for the ovarian cancer [35], the plasma osteopontin, for non-small cell lung cancer [36], and the carcinoembryonic antigen (CEA) for different types of lung cancer [37]. This suggests that computing may be coupled to communication to achieve large sensitivity and accuracy in identifying the cancer type and stage and infer on the precise location.

The typical concentration of tumor cells in the early stages of a disease, or relapse of it, makes their detection from a fixed receiving point extremely difficult. For this reason, as mentioned above, a fundamental component of our proposal, inspired by the state of the art in the relevant technologies, consists of making use of mobile nano-sensors/detectors free to move within the blood.

2.1 Mobile to mobile nano-communications

The detected information has to be delivered to a probe.

According to the biomarkers detected, the subsequent processing has the aim of driving further diagnostic analysis. The capacity of this smart probe of capturing the circulating CTC detectors depends on their concentration. Since the additional number of CTC in the early stage of a disease could be small, even the number of captures could be not enough to be clearly distinguished from the normal concentration in healthy conditions. Thus, it is necessary to use them to stimulate another process having the aim of transferring the available information to the maximum number of particles at the highest speed.

For this purpose, a molecular communication system is needed in order to distribute the point-wise information collected from the sensors to larger particles. The communication system, illustrated in what follows in more details, consists of the emission of a burst of molecules (carriers) by the mobile sensors upon a CTC detection. These carriers propagate within the blood vessel and are hopefully received by all receivers passing through the cloud of the propagating carriers. The objective is to maximize the number of receivers able to receive these propagating carriers. This way, receivers are made aware of the detected CTCs and can carry this information to the smart probe. Clearly, the higher the number of CTCs detected, the higher the alarm level that they can generate.

Since the capture of CTCs is assumed to happen by a mechanism that can be assimilated to a cell particle absorption, the size of the freely moving sensors cannot cause obstruction of vessels. In addition, they can be implemented by using bio-degradable material. In our system, we assume that the smaller sensors, pushed through the blood flow, can detect the cells having their surface populated by the CD47 protein, or other biomarkers, since it can be put in relation with a number of tumors. In fact, it was shown that blocking CD47 can reduce tumor growth by enabling macrophages to eliminate cancer cells [25]. Hence, the antibodies able to block CD47–SIRPα interactions can allow phagocytosis of tumor cells by macrophages [22][23]. In order to allow such treatments, detection of cells exposing the CD47 is of paramount importance. This detection happens through a physical contact between tumor cells and sensors. Detection in sensors can be implemented by exploiting the RNA interference (RNAi) technology [38]. RNAi can be used to implement the detection mechanisms of the CD47 proteins in sensors, by associating it to the antibody of CD47.

The further mechanism exploited by our proposal consists of making use of the so-called bacteriobots [4], as shown in Figure 2. Through the strong attachment of bacteria to Cy5.5-coated polystyrene microbeads due to the high-affinity interaction between biotin and streptavidin, bacteria can drive the attached robot towards the tumor site. The propulsion is provided by flagella, which are cellular appendages, long and thin, arranged in different ways. They have a motor function, and are especially typical of bacillary bacteria. Hence, these bacteria enable the microrobot to move toward tumors, since bacteria preferably move toward tumor cell lysates or spheroids than toward normal cells. In fact, the chemotactic responses of the bacteria to the concentration gradients of lysates or spheroids of solid tumors can determine as the migration of bacteriobots [4].

Hence, robots could implement a combination of microsensor, microactuator, and therapeutic agent. In [4], the microstructure acts as a therapeutic robot containing drugs. In our proposal, they detect specific biomarkers, the presence of which has to be communicated to the external.

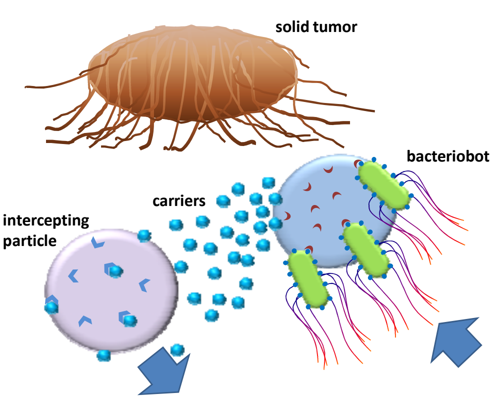

Figure 2. Mobile to mobile nano-communications system.

Since bacteria tend to remain by the tumor cells, it is necessary to use a communication system for transferring the desired information to the intended final recipient. In addition, given the small size of the sensors needed for getting in touch with the tumor site, when they flow through the blood circulatory system, they are pushed towards the endothelium. Since the flow speed at the endothelium is very low, and the number of detections should be (hopefully) small in the early stage of a disease, it follows that the possibility of being detected in a reasonable time (few minutes or even one hour) by a smart probe is very low.

Thus, as show in if Figure 2, we propose to establish a molecular communication between a small, low-speed sensor, which is the transmitter, and many other larger particles, flowing at a significantly larger speed, located close to the longitudinal axis of the blood vessels. This way, the relatively large number of bigger particles, which have a size similar to the white blood cells, can spread the information through the body with a concentration significantly larger than that of CTCs, thus making their detection faster and more reliable. Thus, burst of carriers encoding binary messages are released and captured by mobile receivers having the task of delivering the information to the desired recipient.

The mobile sensors, or robots, are assumed to be made of biodegradable components. Their role is to increase the system's sensitivity of identifying CTCs at very low density, when their density is still small enough in order to allow defining a successful treatment. The bio-degradable composition of CTC sensors may be suitably used for avoiding any possible accumulation of sensors in lymph nodes or elsewhere. In particular, after the introduction of a number of sensors/detectors in the blood stream and their usage as CTC detectors, we assume their elimination by injecting the enzyme able to degrade sensors. The subsequent actions of antibodies and macrophages will eliminate them from the blood circulatory system.

Both tumor detection mechanisms shown above rely on the message to be delivered to their destination in the mobile receivers whose size allows them moving relatively fast in blood vessels. In fact, it is known that larger particles, such as white blood cells, tends to occupy the region in proximity of the longitudinal axis of the vessel. Differently, smaller particles, such as platelets, are bounced towards the endothelium. CTC are typically large cells, having a size similar to that of white blood cells. For this reason, small sensors/detectors are more suitable to get in touch with the tumor site, whilst larger sensors can get in touch with CTCs more frequently.

The same communication system can serve both the tumor detection mechanisms illustrated above, provided that different messages can be exchanged in order to identify different events. Different messages could either be associated with different carriers, each delivering the information of the detection of a different biomarker, or different codewords, provided that a PAM signal can be successfully exchanged within blood vessels between moving entities. This is indeed an objective of our research. This way, a composite information can be delivered to the smart probe allows achieving an improve estimation reliability.

Hence, the medical research objectives and areas mentioned above can be successfully fertilized by biological nano-communication technologies. This is a quite novel research areas [10][11][12][16][17][18], the development of which requires the combined expertise of ICT and biological researchers.

Each time a sensor detects a biomarker, it updates the stored information by comparing the number and the frequency of detections against a number of thresholds and a decision tree. This way, the current status of the sensor reflects the likelihood of the tumor growth. The key aspect is that this information could regulate/trigger the further release of liposomes-loaded siRNA anti CD47.

It is also worth to note that also the number of larger sensors and robots, which have collected the desired information, could be quite small. Thus, also in this case, it is necessary to make use of a molecular communication also for replicating the desired information over a large number of fast moving particles, flowing close to the longitudinal axis of the vessel. This way, we aim at increasing the overall number of cells, flowing at the maximum speed, carrying the desired information to the smart probe.

2.2 Fixed to Fixed nano-communications

Thus, the fast moving particles flow through the circulatory system. We can expect that just a small percentage of them happen to pass by the smart probe. When a particle is captured by a receptor of the probe, the stored information is released to it and the particle status is reset. Clearly, due to the expected small number of interceptions the energy associated with the released signal is not enough to drive further actions. For this reason, a capture is used to trigger a local communication, the scope of which is limited to the probe, having the task of triggering a large number of receptors and collect a sufficient energy to make the information available to the external. These receptors are located over receivers that are stick to probe. Their number is sufficiently high that it can be assumed that the inner probe surface is fully covered by them. This information could be delivered to the external through different mechanisms, such as the release from the probe of a substance that generates an allergic reaction on the skin surface, or the release of a substance that is detectable in the infrared bandwidth or by ultrasound.

In any case it is necessary to analyze also the feasibility of a point-to-point digital communication system, established in the vessel where the probe is inserted, from the sensor capture point to the receivers to be activated on the probe surface. This communications happens through the emission of a burst of biodegradable particles, referred once again to as information carriers, which are involved in the blood flow. The objective of the communication system is to activate the maximum number of receivers, in order to enable a reliable following communication with the external. Due to the need of delivering different messages, binary codewords are used. The subsequent analysis will show the feasibility of the proposal. In particular, the possibility of receiving a transmitted binary codeword will be investigated, along with the avoidance of false positives. A preliminary analysis of this communication scheme has been presented in [39].

The analysis shown below in this paper aims to determine the feasibility of both *fixed-to-fixed* and *mobile-to-mobile* molecular communication system. We will model a receiver by a state diagram. We assume that the state of the receiver, for each individual biomarker, is equal to the number of the received carriers in a given time window W. Thus, when a receiver receives a carrier, the state value of the receiver increases, while after W time units since the reception of a carrier the state value of the receiver decreases. The state Th is the absorbing state, and we say the receiver activated. An activated receiver is a receiver that carries the information "tumor detected".

We used threshold Th values higher than 1 in order to limit false detections, which may happen when CTCs are detected in healthy people. They can be regarded as a noise component affecting the detection process. The numerical results shown in what follows will consider different Th values and will determine the number of receivers that can be activated for different burst sizes of the transmitted carriers.

3. MOLECULAR COMMUNICATIONS

A lot of particle exchange between cells happens in living bodies, that could be either directly exploited or used as inspiration for implementing molecular communication systems. For example, platelets and endothelial cell are terminal points of frequent exchanges of proteins, known as CD40L [15]. Platelets release these protein particles, so that they can bind to the relevant receptors (CD40 receptors) present on the endothelial surface. Typically, this exchange is used to regulate the response to injuries. In fact, by means of this process, platelets activate the endothelial cells, which then recruits monocytes nearby the injury site for neutralizing potential infections.

The communication channel of both this process and similar ones that happen in the bloodstream is quite complex. In what follows, exchanged particles will also be referred to as carriers. The carrier propagation model has to account for the following basic components:

- particle diffusion, modeled as a Brownian motion [9];
- motion drift, caused by the bloodstream, the strength of which depends on the position of the carrier in the vessel. The maximum drift is exerted on the longitudinal axis of the channel, whilst it tends to vanish at the vessel walls. Poiseuille or Casson equations [14] are typically used to model this drift.
- Collisions of the particles propagating in the vessels, both between them and with the endothelium.

Although a number of theoretical diffusion models have already been proposed [9], which also include the drift component [14], a thorough analysis of all the involved components of the propagation model has been published only recently [13]. In particular, it was observed that red blood cells, which are those that most influence the particle motion, due to their concentration and size, push smaller propagating cells, such as platelets, close to the vessel walls. Also white blood cells, which are larger than red ones, contribute to this emerging behavior.

Functional models of a transmitter, a receiver, and a channel between them, and are illustrated in [16]. In this system nodes are assumed to be fixed. The exchanged information is associated with the carrier concentration. Noise components, affecting the receiver capabilities, are analyzed in [10]. The number of the so-called bound receptors, which are receptor-ligand complex pairs, is regarded as the received signal. These papers consider diffusion-based propagation component only, which is not sufficient to suitably model the bloodstream environment.

A different model is analyzed in [11], where the transmitter is assumed to be immersed in a fluid while emitting bursts of carriers, which propagate according to a Brownian motion. The information content is assumed to be associated with the particle release time, according to a pulse position modulation. The situation shown in [17] is rather different. Two communicating entities which exchange molecules propagate through a fluid medium. Their motion model includes both a drift and Brownian component. In [12], the same authors show that the additive inverse Gaussian noise model is suitable for modeling molecular communication channels in fluid media including the drift component. Both an upper and a lower bound of the channel capacity was proposed. Nevertheless, the propagation model used in [12] does not include collisions, which are of extreme importance in our scenario, as it was also observed in [13].

4. COMMUNICATION MODEL

In our analysis we consider blood vessels being at a significant distance from the heart so that it is possible to model the bloodstream without turbulences. Consequently, the macroscopic flow properties, such as velocity and pressure in any position, can be assumed to be constant over time, and the resulting blood motion laminar. As a result of this assumption, the longitudinal shape the vessel can be viewed as a set of concentric cylinders. The space between concentric cylinders is a lamina, and a laminar flow consists of fluid particles moving in longitudinal straight lines of each lamina.

The velocity profile of such laminar flow was shown to have a parabolic shape, modeled by the well-known Hagen–Poiseuille equation, derived from the Navier-Stokes equations [26]-[28]:

$$v(r) = \frac{1}{4\mu}\frac{\Delta P}{L}\left(R^2 - r^2\right) \tag{1}$$

where r is the transversal distance from the longitudinal axis of the vessel, $v(r)$ is the velocity profile, R is the radius of the vessel, μ is the fluid viscosity, and ΔP is the decrease of the pressure along a vessel section of length L. Table 1 reports some known parameters for a blood vessel, used in our simulation.

When large vessels are considered, observations reveal that the velocity profile should be adapted by introducing zero radial velocity gradient, as included in the Casson profile [13]. This model generalizes the motion model of particles in a fluid, as it happens in blood vessels.

The drag force F_d may be modeled by the Stokes law [28] and can be applied to particles having a small Reynolds number (such as red blood cells) in a continuous viscous fluid [29]:

$$F_d = 6\pi\mu R v_p \tag{2}$$

Table 1. Parameter values used in the numerical analysis.

Bacteriobot nodes		General parameters	
Radius	1.0 μm	Vessel length	4.00 mm
Carrier type 1 burst	5000	Vessel diameter	60 μm
Carrier type 1 radius	1.75 nm	Mean flow velocity	0.5 mm/s
Red blood cells (RBC)		Viscosity	1.3 mPa×s
Conc.	4×10^6 U/mm^{-3}	Temperature	310° K
Radius	3.5 μm	Rest. coeff. of vessel wall	0.6
Mobile relay particles		Rest. coeff. of particles collisions	0.9
Conc.	4×10^3 U/mm^{-3}	Simulation time step	100 μs
Radius	5 μm	**Fixed RXs on smart probe**	
Receptor type 1 radius	8 nm	Side (square shape)	14.5 μm
# receptor type 1	10000	Receptor type 2 radius	8 nm
Carrier type 2 radius	1.75 nm	# receptor type 2	10000
Carrier type 2 burst	5000	# fixed RXs	3055

where v_p is the relative velocity of the particle with respect to the flow, given by (1). Through this drag force value, it is possible to estimate both acceleration and velocity of particles. Moreover, for what concerns the diffusion component, as mention above it is modeled as Brownian diffusion, with a coefficient D_m of a quiescent fluid [30]:

$$D_m = \frac{k_B T}{6\pi\mu a} \tag{3}$$

where k_B is Boltzmann constant, a is particles radius, and T is the absolute temperature.

Since the Reynolds number for nano particles is high, equation (2) cannot be applied. It is necessary to consider both convection and diffusion effects, governed by the following equation [14]:

$$\frac{\partial C}{\partial t} + v \cdot \nabla C = D_m \nabla^2 C \tag{4}$$

where C is the particle concentration, D_m is the diffusion coefficient, and v is the fluid velocity vector. An approach alternative to solving equation (4) has been proposed in 1950s and further investigated in [14] and [31]. It consists of introducing a longitudinal diffusion coefficient D_{eff} along the propagation direction. A recent study shown in [13], handles the presence of blood cells by a numerical analysis validated by experimental results. It is shown that such cells can significantly influence the nano-particle propagation and assimilations, so that it is proposed to include them explicitly in the model [13]. What happens is that larger and heavier cells move generally close to the longitudinal vessel axis, whilst the smaller blood components, such as platelets, when collide with heavier blood cells are pushed towards the endothelium. Thus, they remain confined close to walls since they find a less obstructed propagation path. The collisions between particles or between particles and endothelium, can be modeled as inelastic, by using the values of the coefficient of restitution reported in [21]. Thus, our simulation analysis, shown in what follows, integrates the results shown in [13] by modeling collisions between nano particles and both white and red blood cells. This is a significant contribution for suitably addressing the propagation of the information carriers.

4.1 Transmission and reception

The general expression of the bit stream transmitted by the hth nonorobot, or sensor, is:

$$b_h(t) = \sum_{i=-\infty}^{\infty} b_i^{(h)} P_{T_b}(t - i T_b), \qquad (5)$$

where $b_i^{(h)} \in \{0,1\}$ is a sequence of i.i.d. symbols and $P_{Tb}(t)$ is a pulse of a time duration equal to T_b, being T_b the symbol time. This part of the model is worth of some additional comments. The transmission medium illustrated above is quite challenging. If we associate a binary symbol with single burst of carriers released in the medium, we can expect a difficult reception, even affected by a significant probability of false positive reception due to circulating carriers either leftover from other transmissions or endemically part of the blood stream. Hence, in order to make communications more robust we have resorted to repeated pulses. If single impulses or repeated ones are transmitted, the information associated with each transmission is binary ON/OFF. Although this elementary form, it is nonetheless important since it can be used to communicate the presence of a tumor associated with the sensed biomarker. Nevertheless, we also explore the possibility if transmitting different binary message trough such a challenging environment. To this aim, in order to make communications we have resorted to an elementary form of spread spectrum transmission. We associate each bit to be transmitted with a spreading sequence, belonging to the Walsh-Hadamard family of binary sequences. Clearly, any other orthogonal spreading sequences can be used. It can be generated by the following recurrence [40]:

$$h_i = \begin{bmatrix} h_{i-1} & h_{i-1} \\ h_{i-1} & \overline{h_{i-1}} \end{bmatrix}, \qquad (6)$$

starting from the initial value h_0=[1], where $\overline{h_{i-1}}$ is the ones' complement of h_{i-1}. In our experiment we have used the sequences $c^{(1)}$=1111 and $c^{(0)}$=1010. Experiments using longer sequences are being evaluated. The Walsh sequences exhibit zero cross-correlation if perfect synchronization is guaranteed. This is not an easy task to be accomplished, but the high attenuation of the signal in the considered environment (i.e. the high percentage of carriers that cannot be received if the receiver is not very close to the transmitter) protects from mutual interference of transmissions. The signal (5) is spread by a signature sequence: the carrier burst emission time is given by

$$c_k(t) = \sum_{p=0}^{\infty} c_p^{(k)} P_{T_c}(t - p T_c), \quad c_p^{(k)} = c_{p+L}^{(k)}, k \in \{0,1\}. \qquad (7)$$

In (7), L denotes the signature sequence period and T_c is duration of the transmission of each binary element of a spreading sequence, often called chip. The chip duration corresponds to the emission time of a burst of carriers, and T_b/T_c=L.

In summary, the transmitted signal can be expressed as

$$d_h(t) = \sum_{i=0}^{\infty} \sum_{p=0}^{\infty} (\overline{b}_i^{(h)} \wedge c_p^{(0)}) \vee (b_i^{(h)} \wedge c_p^{(1)}) P_{T_c}(t - p T_c), \qquad (8).$$

$$c_p^{(k)} = c_{p+L}^{(k)}, k \in \{0,1\}$$

where the symbols $\wedge$ and $\vee$ indicates the logical *and* and *or* operators, respectively. If we call $r_h(t)$ the corresponding received signal and t_r the receiver synchronization time, that is the time when the receiver detects a received signal, the receiving operations consist of two steps. The first step is a hard estimate of the binary value of the L subsequent samples of the received signal samples, after t_r, collected by using a sampling perio T_c, reffered to as r_h, h=1,...,L, that is $r_1 = r_h(t_r), r_2(t_r - 2T_c), \ldots, r_L(t_r - LT_c)$.

The corresponding estimate of any ith transmitted symbol is realized by a binary to antipodal conversion of all symbol and a subsequent correlation with the used spreading sequences, as follows:

$$\hat{b}_i^h = \underset{k \in \{0,1\}}{\arg\max} \left(\sum_{j=1}^{L} (1 - 2r_h)(1 - 2c_j^{(k)}) \right). \qquad (9)$$

5. PERFORMANCE EVALUATION

5.1 The BiNS2 simulator

The molecular communication system depicted above has been simulated by using the BINS2 simulator. This section illustrates the main functions available in the BiNS2 simulator, in particular those relevant to the considered communication environment; the interested reader can find much more additional details in [6][7][8].

BiNS2 allows modeling biological entities. They can be the communicating nodes transmitting or receiving carriers, or other nodes affecting the signal propagation in a molecular-based communication. It is possible to model the basic microscopic and macroscopic properties of the communication channel, which is the blood stream for our analysis, in some details.

The simulator is implemented in Java. The classes of the package, such as those relevant to nodes and carriers, implement and integrate a general abstract class, named Nano Object. Although they share general features, they also expose different functions. This way, for any simulated scenario it is possible to differentiate each node object type at any time, thus obtaining a multitude of different node objects. The same properties are available for modeling carriers as Nano Objects.

Simulations proceed in discrete time steps, each including different phases, in which objects execute the operations relevant to their specific behavior. The most significant phases are:

- transmission, which implements a carrier emission;
- reception, which implements a carrier assimilation;
- information processing, the behavior of which depends on the received signal intensity, corresponding to the number of received carriers in a time step;
- motion, which implements the mobility models of the simulated biological entities;
- object destruction, for removing objects due to either lifetime expiration or their exit from the region of interest;
- collision management, which includes a collision check phase and a relocation phase, used to locate nodes and carriers after an (in)elastic collision.

A collision can result in either a bounce or an assimilation. The latter happens when a carrier hits a compliant receptor, located on a node surface. Thus, their receiving capacity depends on the number of deployed receptors (see Table 1).

The simulated spatial region can be either unbounded or bounded by a configurable surface. Since in this paper the simulated environment is a blood vessel, we have bounded the simulated region by a cylindrical surface, whose walls are partially inelastic since covered by endothelium (see Table 1).

5.2 Numerical results

Numerical results are relevant to the two communication scenarios presented above:

- *mobile to mobile (m2m) communications*, i.e. communications from bacteriobots to larger relay particles (see Figure 2), which in turn will deliver the information to the smart probe. We assume that after having received a message from a bacteriobot, the mobile particle becomes activated, that is it expresses on its surface a number of specific receptors. These receptors will make it able to bind to the proteins which are expressed on the surface of the smart probe, and thus enable the *fixed to fixed* communication;
- *fixed to fixed communications (f2f)*, i.e. the communication from a relay particle captured by the smart probe to a fixed receivers deployed on the surface of the smart probe. We assume that the bond between an activated mobile particle and the proteins expressed on the smart probe surface is able to hold the mobile device down for a time sufficient to establish a reliable communication [43][44]. Then, the captured particle is triggered to release a message, which can be more complex than the one transmitted by bacteriobots. For instance, it could include not only the type of messages received by a bacteriobot (e.g. the tumor type), but also the number of received messages. The receivers deployed on the surface of the smart probe can decode this message and store it, for subsequent operations.

For what concerns m2m communications, we have considered the transmission of a single burst of 5000 type 1 carriers from a bacteriobot, located on a peripheral position. We have determined the number of mobile particles able to receive carriers. Simulation parameters are reported in Table 1. The number of potential targets, which are present in the simulation space, are equal to about 1600 units in 160,000 time steps, with a time step duration of 100 µs, selected according to criteria stated in [46], hence a simulated time of 16 s. They are continuously created at the left end of the vessel, and destroyed when they get to the right end, with a rate so that the average density reported in the Table 1 is maintained. Thus, the total number of activated targets is a function of the simulation time. Figure 3 shows the projection of the centers of the mobile particles that have assimilated carriers over a transversal section of the vessel in a polar plot, for two different runs. A blue asterisk shows each assimilation. The red circle indicates the position of the transmitting bacteriobot. Since a part of the cloud of carriers is drag away by the blood flow, assimilations are possible in different locations, with a sort of symmetry with respect to the longitudinal axis of the vessel. It may happen that either the cloud of carriers reaches the center of the vessel (Figure 3.a), or its diffusion towards the center is blocked by red blood cells, which flow at higher speed, and carriers are scattered away in a circular fashion (Figure 3.b).

To get a clearer picture of the drag phenomenon, Figure 4.a shows the number of assimilations per mobile target versus its longitudinal distance from the emitting bacteriobot. The ordinate of each asterisk indicates the number of assimilations of a specific mobile particle, whereas the corresponding abscissa value is the distance from the transmitter, averaged at the time of each assimilation. The fitting line shows an expected trend: the larger the distance from the transmitter, the lower the average number of carriers assimilated by mobile receivers, i.e. the received signal strength. This can be easily explained by considering not only the effect of the distance on the diffusion [9], but also the drag effect of the blood flow on both the signal and the receivers. Figure 4.b shows the *delay spread* for each mobile particle able to receive a significant signal, i.e. for each particle able to assimilate a number of carriers larger than a threshold *Th*, set equal to 4. The delay spread is defined as the time interval between the reception of the first and the last assimilated carrier. It appears that the delay spread is nearly constant with the longitudinal distance from the transmitting bacteriobot, up to values in the order of 4 seconds. From the abscissa values, it is also possible to evaluate how many carries are assimilated by any specific mobile particle. In addition, most of the delay spread values are larger than 2 seconds, and all of them are larger than 1 s. We can conclude that, for the specific characteristic of the environment, the only possible type of communication for the m2m scenario is the release of a single, strong pulse of carriers. Different types of message can be encoded with molecular shift keying techniques [41][42], even if, in this way, the number of information which can be delivered is quite limited.

We now analyze the f2f communication system. In this case, even if the drag effect due to the flow is still present, its effect on the carriers close to the wall surface is less important. We obtained the simulation results shown by assuming that the mobile transmitter remains stick to the smart probe surface for the time necessary to receive the command of transmitting its stored information. Thus, it moves during the transmission of the bursts associated to the sequences 1111 and 1010, illustrated in section 4.1. If the transmitter remains stick to the probe surface for all the time needed to complete the transmission of the sequences [43][44], the effectiveness of the communication system would clearly improve. Due to space restrictions, the relevant results are not shown.

The simulated burst size is once again 5000 carriers. We assume to use different, type 2 carriers, but with the same size and weight of those used in m2m communications. The results of our simulation are shown in Figure 5, which illustrates the average number of fixed receivers deployed on the surface of the smart probe able to decode the binary sequence 1111 or 1010, as a function of the threshold value *Th*. The chip duration has been set equal to 1 s after a carefully analysis of reception time. Values for *Th*=1 are not shown for improving the neatness of the figure, also considering that very small threshold values are unsuitable since they are prone to false positives. The preliminary results of Figure 5 are encouraging; for increasing values of the decoding threshold, the number of receivers which decode a wrong sequence, and thus a wrong bit, shrink to zero (*Th*=4), whilst the average number of those able to successfully decode the sequence is nearly one, thus also increasing the robustness of the proposed communication system.

6. DISCUSSION AND FUTURE WORK

In this paper we have analyzed the possibility of implementing molecular communications in blood vessels between fixed and mobile entities. The rationale of this analysis is the use of smart sensors in blood stream are at the core of future medicine with a large number of parameters being monitored. Sensors could then coupled with genetically-modified non-toxic bacteria which move inside tissues or blood with flagella, and detect and accumulate cancer evidences for diagnostic and prognostic purposes by activating smart drugs (see also [4]).

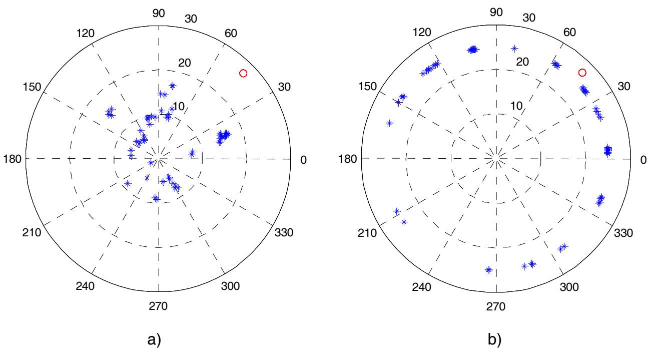

Figure 3. Projection of the centers of the mobile particles that have assimilated carriers over a transversal section of the vessel, for two different simulation runs. Single burst of 5000 type 1 carriers. The red circle identifies the transmitter.

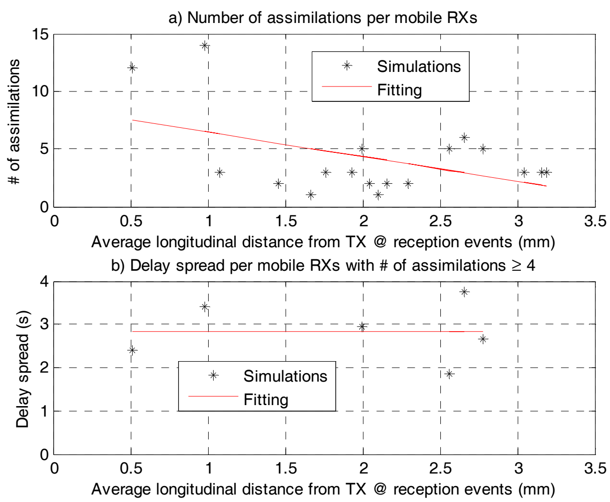

Figure 4. Assimilations (a) and delay spread (b) vs. average longitudinal distance of mobile particles to the bacteriobot.

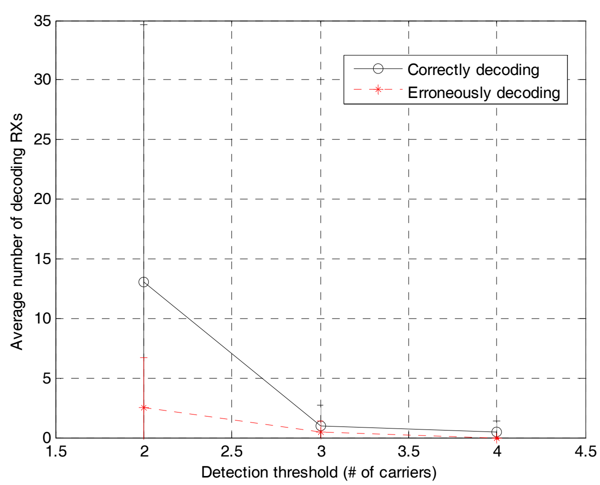

Figure 5. Average number of fixed receivers able to decode the transmitted symbol vs. the detection threshold *Th*. 90% confidence intervals are shown.

It is noteworthy that smart drugs are potent drugs that activate only if certain environmental conditions hold. Smart drugs will deliver therapies on the bases of smart sensors space and time precise information. Therefore they could be more effective than directly targeting the cancer cells with liposomes or soluble proteins.

As an example, recent works report on injected human blood samples, and later mice, with two proteins: E-selectin (an adhesive) and TRAIL (Tumor Necrosis Factor Related Apoptosis-Inducing Ligand). The TRAIL protein bound to the E-selectin protein was able to attach to lymphocytes. When a cancer cell comes into contact with TRAIL, the cancer cell enters apoptosis program and suicide itself [45].

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